

Functional and differential equations

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Functional equations: the toolkit

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Theoretical minimum.

Fact 1 (the catalogue of substitutions). Put $y = 0$, $x = y$, $y = -x$, $x \mapsto \frac{1}{x}$; substitute $f(x)$ for a variable; iterate the equation; compare the equation with itself after swapping x and y . Each substitution is an attempt to extract one of three things: a value of f at a special point, injectivity, or surjectivity — and injectivity is what allows one to cancel f on both sides.

Theorem 1 (Cauchy as the universal reduction). Additivity plus any of the following — continuity at one point, monotonicity on an interval, boundedness above on a set of positive measure — gives $f(x) = cx$; without such an assumption there are non-measurable solutions built from a Hamel basis. The four classical equations reduce to the additive one by taking logarithms or exponentials (proved in Continuous functions – 6).

Theorem 2 (regularity comes out of the equation). An equation in which the unknown appears under an integral is a source of smoothness: from $F(x) = \int_a^x f$ one gets f continuous implies $F \in C^1$, and differentiating the relation raises the regularity by one at each step. The same effect is produced by an equation relating f at points at a fixed distance, and this is what distinguishes a university problem from a school one: half of the work is proving that f is good enough for the formal computation to be legitimate.

Fact 2 (what the answer looks like). For an equation on $\mathbb{N}$ or $\mathbb{Q}$ the answer is forced by induction and no regularity is needed. Over $\mathbb{R}$ the honest answer to “find all f ” is either a one-parameter family plus the remark that continuity is essential, or a proof of non-existence; the standard first line of

a non-existence proof is to derive a contradiction from monotonicity and a growth estimate.

1. (IMC) Let $\mathbb{R}_+$ be the set of positive reals. Find all functions $f: \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $f(x)f(yf(x)) = f(x+y)$ for all $x, y \in \mathbb{R}_+$.
2. (IMC) Prove that there is no function $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(0) > 0$ such that $f(x+y) \geq f(x) + yf(f(x))$ for all $x, y \in \mathbb{R}$.
3. (RMO 2014) Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that
 - (i) $\text{id} + f$ is non-decreasing;
 - (ii) for some natural m the function $\text{id} + f + f^2 + \cdots + f^m$ is non-increasing,
 where the exponent denotes the functional power.
4. (RMO) Find all continuous functions $f: \mathbb{R} \rightarrow [1, \infty)$ for which there are $a \in \mathbb{R}$ and a positive integer k with $f(x)f(2x) \cdots f(nx) \leq an^k$ for all real x and all positive integers n .
5. Solve the Cauchy equation $f(x+y) = f(x) + f(y)$ on $\mathbb{Q}$ without any regularity, prove that a monotone solution on $\mathbb{R}$ is linear, and construct a non-measurable solution from a Hamel basis; explain precisely which step of the school argument breaks over $\mathbb{R}$.